

Physics

Middletown High School | Grade 12, Fall Semester 2026

Teacher: Mr. D. Oyelaran
Room: 108 Lab
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Course Description

An algebra based physics course built around laboratory work. Students measure motion, forces and energy, then use the data to test the models that describe them.

Course Units

Weeks 1 to 3	Ch1: Kinematics Motion without reference to its causes. Key equations $v = u + at$, $s = ut + 0.5at^2$ and $v^2 = u^2 + 2as$. Projectile motion as uniform horizontal motion plus vertical free fall.
Weeks 4 to 6	Ch2: Newton's Laws Inertia, $F = ma$, and equal and opposite reaction. Free body diagrams are required on all problems in this unit.
Weeks 7 to 9	Ch3: Work, Energy and Power Work as force times displacement, kinetic and potential energy, conservation of energy, and power as the rate of doing work.
Weeks 10 to 12	Ch4: Waves and Light Wave speed, frequency and wavelength, reflection and refraction, and the behavior of light through lenses.
Weeks 13 to 16	Ch5: Electricity and Magnetism Current, voltage and resistance, Ohm's law, series and parallel circuits, and magnetic fields around a current.

Key Dates This Semester

Oct 2	Lab Report due: Motion
Oct 22	Mid-Term Exam, covers Ch1 to Ch3
Nov 20	Electricity Project due
Dec 16	Final Exam

School wide dates

Sep 8	First day of classes
Oct 19 to Oct 23	Mid-term examination week
Nov 6	Parent teacher conferences
Nov 25 to Nov 27	Thanksgiving break, no classes
Dec 14 to Dec 18	Final examination week

Materials

Scientific calculator, lab notebook, safety goggles (provided, kept in the lab), ruler.

Grading Policy

25 percent	Laboratory work and reports
15 percent	Homework
15 percent	Quizzes
25 percent	Unit tests
20 percent	Final exam

A = 90 to 100 percent, B = 80 to 89, C = 70 to 79, D = 60 to 69, F = 59 and below.

Late and Make-Up Work

Work handed in late without an excused absence receives a maximum of 50 percent credit. With an excused absence, work is due within one week of returning. It is the student's responsibility to ask what was missed. Make-up tests are held on Thursdays after school and must be booked at least one day ahead.

Academic Honesty

All work submitted must be your own. Copying another student's work, or submitting text or code generated by someone or something else as your own, is treated as academic dishonesty and results in a zero for that assignment and a call home.

Extra Help

Extra help is available Tuesdays and Thursdays, 3:15 to 4:00 pm, in the room listed above.